

Isabella Gilarranz

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EDUCATION

University of Miami <i>B.S. Electrical Engineering & B.S. Computer Science, Minor in Mathematics GPA: 3.4</i> Relevant Coursework: Data Structures & Algorithms, Theory of Computation, Introduction to AI, Statistical Analysis Honors: President's Scholarship Recipient Leadership & Activities: Stanford BEHAVIOR Challenge 2026 Participant Member, American Society of Mechanical Engineers (ASME) Member, UMaker	Coral Gables, FL Expected August 2027
Broward College — Robert “Bob” Elmore Honors College <i>A.A. Computer Science</i> Honors: High Honors Graduate Leadership & Activities: President, STEM Club VP of Leadership, Phi Theta Kappa Honor Society	Pembroke Pines, FL 2023 – 2024

EXPERIENCE

Undergraduate Researcher <i>University of Miami SEBTRAC / Industrial Assessment Center</i> - Analyze building performance, HVAC operations, and field data to identify energy-efficiency opportunities and support AI-powered energy management research. - Quantify potential energy and cost savings from operational improvements, translating technical findings into data-driven recommendations for building performance.	Aug 2026 – Present Coral Gables, FL
Undergraduate Researcher <i>University of Miami Robotics and Mechatronics Lab</i> - Developed a Python-based autonomous planning system for the BEHAVIOR Challenge 2026, enabling successful plans across all 100 benchmark tasks. - Built automated testing and regression checks to improve system reliability and support integration with the team's downstream robotics software. - Lead the end-to-end build and systems integration of a 22-actuator humanoid robot, coordinating mechanical fabrication, electronics, wiring, and actuator systems.	May 2026 – Present Coral Gables, FL
Data Analyst, Business Intelligence <i>Safra National Bank of New York</i> - Founding analyst on the bank's Business Intelligence team, created to modernize reporting from legacy data systems. Built its first 7 Power BI and Tableau dashboards, enabling self-service reporting across 10 of 12 departments. - Built the SQL data layer behind the dashboards, transforming a decade of banking data (2014–2025) into query-ready models and defining KPIs with department managers to address real business needs.	Dec 2024 – May 2025 Brickell, FL
IT Help Desk Intern <i>Safra National Bank of New York</i> Diagnosed and resolved hardware and software issues across end-user workstations, collaborating with network administrators and vendors to restore system functionality.	May – Dec 2024 Brickell, FL
Brain-Machine Interface Research Intern <i>SBSHS Professional Enhancement Program</i> Analyzed EEG signals from mouse behavioral experiments in Python and presented findings at the Annual Pillars Conference.	Feb – May 2024 Davie, FL

PROJECTS

MotionSense_IMU <i>KiCad, USB-C, I²C</i> - Designed and fabricated a custom USB-C-powered 6-axis IMU sensor board in KiCad, including power regulation, I²C communication, and supporting circuitry. - Completed schematic capture, PCB layout, design verification, and manufacturing through JLCPCB; currently performing hardware bring-up and ESP32 driver development.	2026
Portable MP3 Player <i>ESP32, C++, I2S</i> Designed and built a working handheld MP3 player from scratch: I2S digital audio to a DAC, TFT display UI, SD card storage, and tactile controls, writing all firmware in C/C++; battery power and a custom KiCad PCB revision in progress.	2026

TECHNICAL SKILLS

Programming & Data: Python, SQL, C/C++, Java, Git, Power BI, Tableau, Excel
AI & Software: AI Planning, Fast Downward, PDDL/BDDL, CI / Testing
Hardware & Embedded: KiCad, PCB schematic capture & layout, ESP32, I²C, SPI, I2S
Lab Equipment: Oscilloscope, logic analyzer, bench power supply, digital multimeter, soldering station
Languages: Fluent English & Spanish